

Professional Summary

Result-oriented Graduate Researcher and incoming PhD student in Research Methods and Statistics (RMS) at the University of Denver, with a record of academic excellence and community leadership. Proven ability to balance rigorous methodological analysis with high-level program coordination, facility logistics, and student mentorship. Highly dependable and punctual, with extensive experience managing complex data projects and supporting diverse populations in both academic research and front-line service environments. Seeking to leverage strong organizational skills, analytical expertise, and a commitment to student success to contribute effectively to the DU campus community through research, teaching, or administrative support.

Core Competencies: Advanced Statistical Modeling; Data Science & Predictive Analytics; Quantitative & Qualitative Research Methods; Mathematical Modeling; Machine Learning (R, Python, SAS); Spatio-temporal Analysis; Survey Design & Program Evaluation; Data Visualization & Interpretation.

Education

- Sept 2026 **PhD in Research Methods and Statistics (RMS)**, *University of Denver*, Denver, CO
(Expected) Incoming Doctoral Student – Morgridge College of Education
- Jan 2025 - **M.S. Mathematical Data Science**, *Tarleton State University*, Tx, USA, CGPA: 3.8/4.00
- May 2026 Capstone Title: Hybrid Bayesian-Ensemble Framework for Early Detection of Perinatal Depression. Advisors: Prof. Scott Cook & Prof. Jesse Crawford
- Jan 2021 - **MPhil Mathematical Statistics**, *Kwame Nkrumah University of Science and Technology*, Ghana,
Nov 2025 CGPA: 3.28/4.00
Thesis Title: Misspecifications of SARIMA Time Series Models: Simulation-Based Study. Advisor: Prof. Sampson Twumasi-Ankrah (PhD)
- Aug 2015 – **B.Sc. in Statistics**, *Kwame Nkrumah University of Science and Technology*, Ghana, CGPA:
July 2019 3.40/4.00
Thesis Title: Time Series Modelling of Road Traffic Accidents in the Kumasi Metropolis, Ghana. Advisor: Prof. Nana Kena Frimpong (PhD)

Honors and Awards

- 2nd Best Poster Presenter - Life, Health and Agricultural Sciences Division, 2026 RIED Research Symposium, Tarleton State University, Tx, USA (March 2026)
- Davidson Presidential Scholarship, Tarleton State University, TX, USA (Prestigious award recognizing academic excellence and leadership)
- Lozano Long Division of Global, Community, and First Gen. Scholarship, Tarleton State University, TX, USA (Award supporting global perspectives, community engagement, and first-generation students)
- Pentecost Brilliant but Needy Scholars Award, Ghana (Recognition for academic brilliance and financial need support during undergraduate studies at KNUST)
- 3rd Best Poster Presenter, UDS Annual Graduate School Conference, Ghana (November 2021)

Research & Data Analytics Experience

- Fall 2025-Spring 2026 **Predictive Models for Early Detection of Postpartum Depression**, *Capstone Project*, *Tarleton State University*, TX, USA, Advisors: Prof. Jesse Crawford & Prof. Scott Cook
2026 Developed a Hybrid Bayesian-Ensemble Framework for early detection of perinatal depression using 14,008 records. Optimized an XGBoost ensemble ($R^2 = 0.723$) and integrated a PyMC Bayesian GLM for uncertainty quantification (94% HDI). Utilized SHAP for model interpretability to provide clinical decision support.
- Summer 2025 – May 2026 **Volunteer Graduate Research Assistant**, *Medical Laboratory Sciences*, *Tarleton State University*, TX, USA, PI: Gandhi Subi (PhD)
Supports data analysis using RStudio and Python to study COVID-19's impact on mental health. Contributes to survey design, data cleaning, and statistical interpretation of behavioral health outcomes.

- Summer 2025 **Graduate Research Assistant – RISER Project**, *Dept. of Health Education and Rehabilitation Services, Tarleton State University, TX, USA*
Led data analysis using RStudio and Qualtrics for community-based health research. Designed surveys with IRB-approved protocols, ensuring data security and integrity. Collaborated on grant writing and community outreach planning.
- Fall 2025 **Mathematical Modeling of Mayaro Virus Transmission**, *MATH 5330 Final Project, Tarleton State University, TX, USA*
Constructed and analyzed deterministic and stochastic SEIR-type models to simulate virus transmission dynamics. Developed visualizations to communicate infection risks and potential intervention strategies for public health authorities.
- 2020 – 2024 **Research & Field Assistant (Public Health focus)**, *Ghana Health Service, ISSER, and others, Ghana*
Supported national projects in public health and behavioral science. Conducted quantitative data collection (surveys, KIIs, FGDs). Managed data using REDCap, STATA, and Excel, ensuring high data quality for large-scale social development studies.

Professional Experience

- Aug 2024 – **Biostatistician**, *Kintampo Health Research Centre, Kintampo, Ghana*
Jan 2025 Designed statistical models (regression, survival analysis) for health studies. Managed datasets using R, Stata, and SPSS. Collaborated with epidemiologists and policymakers to inform interventions. Authored reports and manuscripts.
- Apr 2021 – **Biostatistician**, *Center for Health Systems Strengthening, Kumasi, Ghana*
Nov 2021 Performed analyses using Stata and R on clinical and survey data. Developed statistical methods and prepared reports for conferences.
- May 2022 – **Reviewer**, *Jozac Publishers, South Africa*
Dec 2024 Reviewed manuscripts for statistical rigor and methodological soundness.

Publications, Manuscripts Under Review, & Presentations

1. Publications

- Annan V, Boateng EA, Ofori S, **Keku E**, Dotse BP, Zuuri CN. Lived experiences of individuals affected by HIV/AIDS: understanding the dynamics of stigma and discrimination within the Sunyani Municipality, Ghana. *BMJ Public Health*, 2026;4:e003806. <https://doi.org/10.1136/bmjph-2025-003806>
- Quantson M, **Keku E**, Kwamekyi A, Quantson A, Aruna S. (2025). Rural inhabitants' perception-induced immunisation status: The case of yellow fever vaccination in the Ashanti Region of Ghana. *Rwanda Journal of Medicine and Health Sciences*, 8(2), 514–529. <https://dx.doi.org/10.4314/rjmhs.v8i3.6>
- **Keku, E.** (2025). God hears. Do we move? Rethinking prayer, progress, and health (PPH) in Ghana. *Christian Journal for Global Health*, 12(2). <https://doi.org/10.15566/cjgh.v12i2.483>
- **Keku, E.**, et al. (2024). Prevalence of postpartum depression in Ghana: A systematic review study. *Scientific Electronic Archives*, 17(1). <https://doi.org/10.36560/17120241826>
- **Keku, E.**, et al. (2022). The extent of self-reported impact of COVID-19 on daily academic activities and incomes among university students in Kumasi, Ghana. *African Social Science and Humanities Journal*, 3(4), 129–136.
- Babalola, E. O., & **Keku, E.** (2024). Ethno-STEM integrated project-based learning to improve students' creative thinking skills. *International Journal of Ethnoscience and Technology in Education*, 1(2), 116–130. <https://doi.org/10.33394/ijete.v1i2.11308>
- **Keku, E.**, et al. (2023). Prepaid meter issues in Ghana: Assessing the impact and the way forward. *Scientific Electronic Archives*, 16(4). <https://doi.org/10.36560/16420231695>
- **Keku, E.**, & Patterson, E. (2022). The impact of social media outages on Ghanaian users: A snapshot of the October 2021 Facebook outage event. *Academic Voices*, 15–18.
- Osei, E. A., **Keku, E.**, Aluah, D., & Okoro, J. P. (2023). Perceptions on farming as a profession: Teenagers under the spotlight in Kwakyeokrom in the Nsawam-Adoagyiri Municipality, Ghana. *Scientific Electronic Archives*, 16(1). <https://doi.org/10.36560/16120231653>

2. Manuscripts Under Review

- **Keku, E.** COVID-19 vaccine acceptability assessment in Ghana. *Pan African Medical Journal*.
- **Keku, E.** Lived experiences of individuals affected by HIV/AIDS: Understanding the dynamics of stigma and discrimination. *Journal of the International AIDS Society*.
- **Keku, E.** Adverse Birth Outcomes and Associated Factors Among Adolescent Mothers in the Tolon District.

- **Keku, E.** Prevalence and Associated Factors of Postpartum Anaemia Among Adolescent Mothers in the Tolon District. *Journal of Pregnancy.*

3. Presentations

- **Keku, E.** (March 2025). Ethno-STEM Integrated Project-Based Learning to Improve Students' Creative Thinking Skills. Oral presentation at the 104th Mathematical Association of America Conference, Prairie View A&M University, USA.
- **Keku, E.** (April 2025). An Effect of Climate Change and Health Outcomes in Ghana: A Systematic Review Study. 3 Minute Challenge Oral Presentation, Tarleton State University, USA.
- **Keku, E., et al.** (2021–2022). The extent of COVID-19 impact on daily activities and income among university students in Kumasi, Ghana.
 - Poster presentation at UDS Annual Graduate School Conference, Ghana (2021).
 - Oral presentation at the 5th Annual Africa Interdisciplinary Health Conference, Ghana (2022).

Teaching Experience

- Jan 2025 – **Graduate Teaching Assistant**, *Tarleton State University*, Stephenville, TX, USA
- May 2026 Taught undergraduate courses in College Algebra and Introductory Statistics, preparing lectures and assessments. Provided tutoring in Pre-Calculus and Calculus I–III via the Math Achievement Center. Conducted classroom observations with faculty feedback and participated in TEA seminars to enhance inclusive teaching strategies. Assisted with grading and curriculum development.
- Sep 2019 – **Undergraduate Teaching Assistant**, *Department of Epidemiology and Biostatistics, KNUST*,
Aug 2020 Ghana
Assisted faculty in public health and biostatistics instruction. Facilitated tutorials and workshops, achieving over 90% student success. Simultaneously supported faculty research through high-volume data entry, cleaning, questionnaire digitization, codebook development, and preliminary descriptive analyses for multiple epidemiological studies using REDCap, Excel, and STATA.
- Oct 2014 – **Academic Tutor**, *New Fountain Erudite School*, Accra, Ghana
- Aug 2015 Taught mathematics and design and technology to junior high students. Adapted lesson plans, evaluated performance via quizzes, and maintained accurate records.

Conferences/Workshops Attended

- 105th Mathematical Association of America Conference, Baylor University, TX (March 2026).
- Conference for the Advancement of Mathematics Teaching (CAMT), San Antonio, TX (July 2025).
- 104th Mathematical Association of America Conference, Prairie View A&M University, TX (March 2025).
- "AI Era Research Integrity: AMA with Paperpal Preflight," Paperpal Preflight (May 2024).
- Webinar: "How to Get More Readership and Citations," Kudos (virtual; May 2023).
- "Model Assists Users {ChatGPT3, Quillbot}," Prof. Funmilayo Adesope, Nigeria (virtual; May 2023).
- Monitoring and Evaluation Webinar, Capacity Africa Institute, Kenya (virtual; November 2022).
- 5th Annual Africa Interdisciplinary Health Conference, Ghana (August 2022).
- 2nd Annual Applied Research Conference, Accra Technical University, Ghana (July 2022).
- UDS Annual Graduate School Conference (November 2021).
- 4th Annual Africa Interdisciplinary Health Conference, Rwanda (virtual; August 2021).
- Integrity Leadership Training Workshop, International Leadership Foundation (ILF) (AY 2015/2016).

Skills and Competencies

Technical Skills: R (Advanced); Python (Pandas, Scikit-learn, PyMC); SQL; SQL Server; Power BI; Tableau; Stata; SPSS; LaTeX; Excel (Advanced); REDCap; Qualtrics.

Professional Strengths: Data Visualization; Statistical Reporting; IRB Compliance; Collaborative Problem Solving; Program Evaluation; Quantitative Research Design.

Volunteering and Leadership Experience

- Risk Manager and Chairperson, Tarleton State University International Student Organization (March 2025 – Present).
- Member, African Health Literacy Network, Ghana (August 2023 – Present).
- Member, Christ Impact Lives Foundation (CILF), Accra, Ghana (July 2022 – Present): Aided needy single

mothers and young widows.

- Member, Adez Foundation, Kumasi, Ghana (July 2020 – May 2022): Supported orphans and prisoners with educational materials.
- President, West African Federal Hostel, KNUST, Ghana (May 2019).
- Academic Mentor, Department of Statistics, KNUST (September 2017 – May 2019).

References

- **Subi Gandhi, PhD**, Professor & Co-Director, Medical Laboratory Sciences & Nutrition, Tarleton State University, TX, USA; gandhi@tarleton.edu; 254-968-0578
- **Jamie Jackson, PT, DPT**, Clinical Assistant Professor, Physical Therapy Program, Tarleton State University, TX, USA; jjackson1@tarleton.edu; 254-595-7264
- **Triss Ashton, PhD**, Associate Professor, Management Department, Tarleton State University, TX, USA; ashton@tarleton.edu; 325-656-2852
- **Scott Cook, PhD**, Chief Data Scientist, University Strategy, Tarleton State University, TX, USA; scook@tarleton.edu; 254-968-1958
- **Sampson Twumasi Ankrah, PhD**, Associate Professor & Research Consultant, Statistics and Actuarial Department, KNUST, Ghana; stankrah2017@gmail.com; +233 24 497 4531